



# Effect of crystallographic orientation on mechanical anisotropy of selective laser melted Ti-6Al-4V alloy

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## ABSTRACT

The crystallographic texture of Ti-6Al-4V produced by selective laser melting (SLM) under various laser energy densities was characterized by electron backscatter diffraction technique to explore its effect on the anisotropy in tensile properties. Results show that crystallographic orientation depending on laser energy density acts a significant role in determining the mechanical anisotropy of SLMed Ti-6Al-4V samples. The microstructure of the SLMed Ti-6Al-4V samples consists of fully martensites. As for the martensites, the fraction of basal orientations decreases, while the content of prismatic orientations increases with laser energy density increasing from 101 to 269 J/mm<sup>3</sup>. And the order of the dominated crystallographic orientation of martensites with the laser energy density is  $(12\bar{3}0)[2\bar{1}13] \rightarrow (11\bar{2}4)[\bar{1}341] \rightarrow (11\bar{2}0)[\bar{1}101] \rightarrow (11\bar{2}0)[2203]$ . There is anisotropy in tensile properties between horizontally and vertically built samples, which is more obvious with laser energy density. The formation of such anisotropy is ascribed to the higher Schmid factor values of the grains in the vertically built tensile samples than those in horizontally built ones.

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## 1. Introduction

Ti-6Al-4V alloy has been widely used in aerospace, energy, medical and chemical processing industries due to its combination of excellent strength, fracture toughness, low density and high biocompatibility. Recently, selective laser melting (SLM), a powder bed-based additive manufacturing technology, is particularly attractive for the fabrication of Ti-6Al-4V parts with complicated shapes owing to its advantages of a lower time-to-market, a near-net-shape production, a high material utilization rate and a high level of flexibility [1].

Unfortunately, there are always significant microstructural and mechanical anisotropies in SLMed components to restrict their applications [2–4]. Especially, the anisotropies of tensile properties [5–7], ductility [4] and fatigue performance [8] are frequently observed in SLMed Ti-6Al-4V alloy. Undoubtedly, their anisotropies are the result of its special manufacturing conditions and intrinsic material characteristics. On the one hand, the SLMed Ti-6Al-4V alloy is mainly composed of hexagonal closed packed (HCP) martensite  $\alpha'$  with inherent low crystal symmetry [9]. Generally, HCP materials have high sensitivity to anisotropy due to the interactions between crystallographic slip and deformation twinning [10,11]. On the other hand, a steep temperature gradient exists between the

center and edge of the molten pool surface during SLM [12], which is also responsible for such anisotropy. Such directional solidification process of SLM causes the columnar grains to grow preferentially along the well-defined directions, i.e.  $\langle 100 \rangle$  direction is parallel to the maximum heat extraction direction, resulting in a high anisotropy [13].

It is widely accepted that the crystallographic texture (usually controlling slip mode and plastic flow) exhibits a strong link with the mechanical anisotropy of SLMed Ti-6Al-4V alloy [14,15], even though the existence of columnar grains [4] and molten pool behavior [6] would also produce an influence. Thus, it is feasible to eliminate the mechanical anisotropies of SLMed Ti-6Al-4V alloy by optimizing the crystallographic textures. Nowadays, considerable researches have reported that a dominant  $\{100\}$   $\beta$  texture forms in SLMed Ti-6Al-4V alloy, while  $\alpha'$  texture is relatively weak due to the high number of  $\alpha'$  variants within each prior  $\beta$  grain [16–18]. However, there has been little effort until now to establish the relationship between the crystallographic texture and mechanical anisotropy of SLM Ti-6Al-4V alloy. Only, Simonelli proposed that fracture of SLMed Ti-6Al-4V alloy occurred in none of the basal or prismatic planes that are instead generally associated with  $\alpha$  cleavage fracture [4]. It is, therefore, essential to further analyze the type and volume fraction of  $\alpha'$  texture under various SLM processing parameters, as well as its influences on anisotropic mechanical properties.

In the present work, texture measurement and mechanical tests were carried out for SLMed Ti-6Al-4V alloy under various laser energy densities in order to systematically establish their relationships

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between crystallographic textures and anisotropic behavior. In addition, an attempt was made to weaken the mechanical anisotropies by regulating the SLM processing parameters to tailor the crystallographic textures.

## 2. Experimental

All SLM experiments were conducted using a self-developed SLM system (LSNF-I), whose details have been described in our previous publications [19–21]. The starting material was gas-atomized, spherical Ti-6Al-4V powders, and D10, D50 and D90 of the particle size distribution parameters are 18, 27 and 40  $\mu\text{m}$ , respectively (Fig. 1). All samples with dimension of  $10 \times 10 \times 10 \text{ mm}^3$  were deposited on the commercially pure Ti substrate in an argon environment. The applied SLM processing parameters of the samples are given in Table 1.

Then, all the samples were subjected to microstructural and textural characterization by an X'Pert PRO X-ray diffraction (XRD) and a FEI Sirion 200 scanning electron microscope (SEM) equipped with an OIM (TSL) system. The crystallographic orientations on cross and vertical sections of the samples were analyzed by electron backscatter diffraction (EBSD). During EBSD measurement, the step size was reduced to 0.20  $\mu\text{m}$  to obtain more details, and the average image quality of the obtained electron backscattering diffraction pattern was  $>60$ . Also, disorientations  $<2^\circ$  were excluded from the data to improve the accuracy. The measured orientation data were processed by the TSL acquisition and analysis software JTEX [22] to obtain orientation maps as well as orientation distribution functions (ODFs).

Finally, plate-type tensile test pieces were designed according to ASTM E8 standard. A series of SLMed Ti-6Al-4V tensile test pieces under various SLM processing parameters were deposited in their net dog bone shape along two orthogonal orientations (vertical and horizontal), as shown in Fig. 2. The building direction (BD) of the SLMed part is considered to be the angle between the substrate and the longest axis of part. Hence, the BD is  $90^\circ$  and  $0^\circ$  for vertically and horizontally built tensile samples, respectively. The tensile tests were performed on the INSTRON 1341 electro-hydraulic servo testing machine at room temperature. Also, each tensile test value was obtained by mathematically averaging the test results of five SLMed samples under the same SLM processing parameters. Moreover, fracture surfaces of tensile samples were observed using SEM.

**Table 1**

The processing parameters used in the experiments.

| No. | Laser power $P$ , W | Scanning velocity $V$ , mm/s | Hatch spacing $S$ , mm | Layer thickness $T$ , mm | Laser energy density $E_v$ , J/mm <sup>3</sup> |
|-----|---------------------|------------------------------|------------------------|--------------------------|------------------------------------------------|
| 1   | 194                 | 800                          | 0.12                   | 0.02                     | 101                                            |
| 2   | 194                 | 1000                         | 0.08                   | 0.02                     | 121                                            |
| 3   | 194                 | 600                          | 0.07                   | 0.02                     | 230                                            |
| 4   | 194                 | 400                          | 0.09                   | 0.02                     | 269                                            |

## 3. Results and discussion

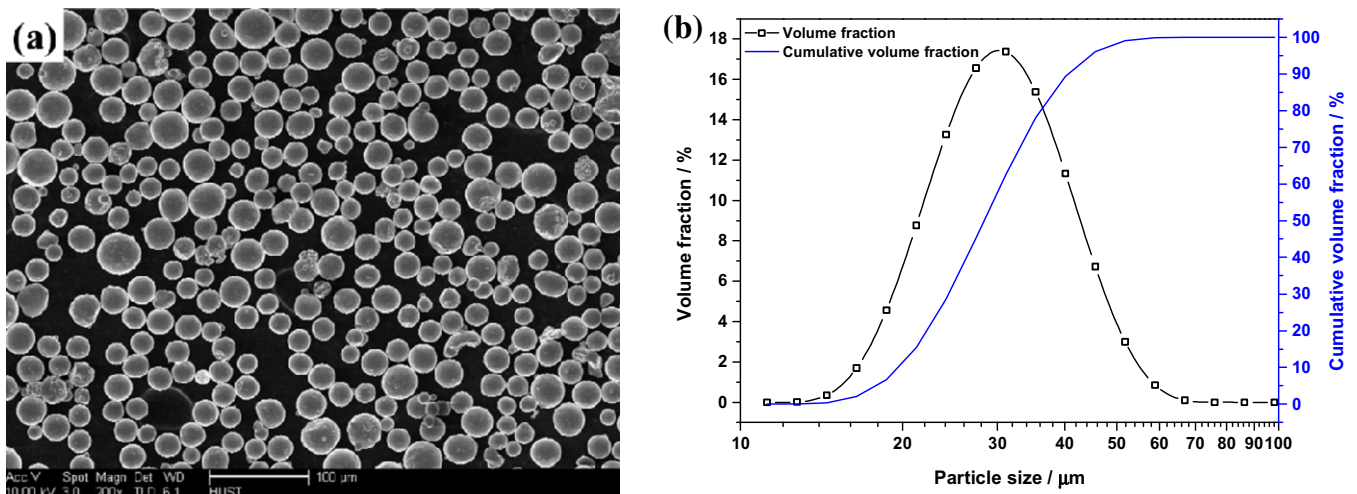
### 3.1. Microstructure

The relative densities of all the SLMed Ti-6Al-4V samples are rather high ( $>99.9\%$ ), implying that there are no obvious defects (pores, crack, etc.). XRD patterns of the SLMed Ti-6Al-4V samples on XY and XZ planes are revealed in Fig. 3. Clearly, the XRD patterns on XZ plane exhibit strong diffraction peaks at  $35^\circ$ ,  $38^\circ$ ,  $40^\circ$ ,  $53^\circ$ ,  $63^\circ$ ,  $71^\circ$ ,  $77^\circ$  and  $78^\circ$ , which are the characteristic of hcp  $\alpha/\alpha'$ -Ti hexagonal. During SLM, the cooling effect is mainly caused by the substrate, and heat mostly dissipates along the negative Z-direction. Therefore, the intensity of XRD peaks between XY and XZ planes is entirely different due to their different thermal and solidification conditions, implying that their crystallographic orientation may be unlike. Besides, there is a small peak at  $\sim 57^\circ$  in  $2\theta$  of XRD profile on the XY plane, indicating that a small amount of  $\beta$  phase is retained in the SLMed samples.

Fig. 4 exhibits the microstructures on XY and XZ planes of the SLMed Ti-6Al-4V sample. Obviously, a distinctive chessboard pattern structure is observed on XY plane due to the applied hatch angle of  $90^\circ$  (Fig. 4(a)), which is the characteristic of SLM. The average width of squares is approximately equal to the applied hatch spacing (80  $\mu\text{m}$ ). Also, the columnar prior  $\beta$  grains filled with fine acicular  $\alpha'$  martensites are seen on XZ plane in Fig. 4(b), which is consistent with the previous literatures [23,24]. These martensites with thickness at the sub-micron scale are parallel or perpendicular to each other, as shown in Fig. 4(c, d).

### 3.2. Crystallographic texture

The image quality photos and orientation images on XY and XZ planes of the SLMed Ti-6Al-4V sample deposited with  $E_v$  of 121 J/mm<sup>3</sup> are exhibited in Fig. 5. Similarly, the grid structure (Fig. 5(a, b)) and columnar grains (Fig. 5(c, d)) are also observed on XY and XZ planes,



**Fig. 1.** The morphology (a) and particle size distribution (b) of the Ti-6Al-4V powders.

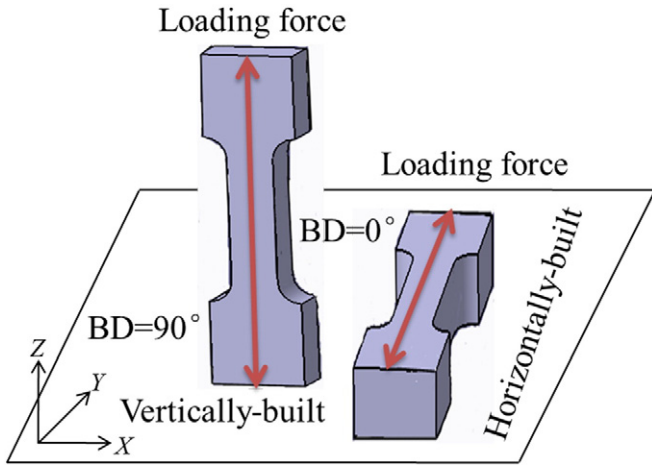


Fig. 2. The build orientation of the vertically and horizontally built SLMed Ti-6Al-4V tensile test pieces using coordinate system defined by ISO/ASTM 52921.

respectively. The existence of various martensitic ( $\alpha'$ ) crystallographic orientations in SLMed Ti-6Al-4V implies a weak texture of  $\alpha'$  due to the relatively high number of variants within each columnar  $\beta$  grain, which is consistent with previous study [16,17]. The  $\varphi_2 = 0\text{--}60^\circ$  ODF sections of SLMed Ti-6Al-4V samples deposited under various laser energy densities are given in Fig. 6. Clearly, the  $\alpha'$  laths of the SLMed components have a weak texture. Although the ODF intensities located at some special orientations remain at a relatively high level, the maximum value is still below 55. Obviously, as the applied laser energy density increases from 101 to 269 J/mm<sup>3</sup>, the dominated crystallographic orientation of martensites varies accordingly in the following order:  $(12\bar{3}0)[2\bar{1}\bar{1}3] \rightarrow (11\bar{2}4)[\bar{1}\bar{3}41] \rightarrow (11\bar{2}0)[1\bar{1}01] \rightarrow (11\bar{2}0)[2\bar{2}03]$ .

According to the calculation, the volume fraction of the orientations corresponding to the main slip systems is below 1% in all the samples, which is too small to produce a significant effect on the properties. Thus, Fig. 7 gives the calculated volume fraction of several other typical types of orientations in SLMed Ti-6Al-4V samples according to ODFs by analysis software JTEX, including  $\{0002\} \langle 1\bar{1}00 \rangle$  basal,  $\{11\bar{2}2\} \langle 0\bar{2}21 \rangle$  pyramidal,  $\{22\bar{4}1\} \langle 1\bar{2}12 \rangle$  pyramidal and  $\{11\bar{2}0\} \langle 1\bar{1}02 \rangle$  prismatic orientations. Also, the content of  $\{0002\} \langle 1\bar{1}00 \rangle$  basal orientation is always less than that

of  $\{11\bar{2}0\} \langle 1\bar{1}02 \rangle$  prismatic orientation in the SLMed Ti-6Al-4V samples. With increasing  $E_V$  from 101 to 121 J/mm<sup>3</sup>, the fraction of  $\{0002\} \langle 1\bar{1}00 \rangle$  basal orientation decreases from 1.97% to 1.36%, while the fraction of  $\{11\bar{2}2\} \langle 0\bar{2}21 \rangle$  pyramidal orientation increases to 2.63%. Further increasing the applied  $E_V$  to 230 J/mm<sup>3</sup>, the fractions of  $\{0002\} \langle 1\bar{1}00 \rangle$  basal and  $\{11\bar{2}2\} \langle 0\bar{2}21 \rangle$  pyramidal orientations decrease to be 1.02% and 1.16% respectively, but  $\{22\bar{4}1\} \langle 1\bar{2}12 \rangle$  pyramidal and  $\{11\bar{2}0\} \langle 1\bar{1}02 \rangle$  prismatic orientations occupy a leading position with the contents of 13.09% and 10.07%, respectively. Furthermore,  $\{11\bar{2}0\} \langle 1\bar{1}02 \rangle$  prismatic orientation becomes the dominant one with the fraction of 13.15% in SLMed Ti-6Al-4V sample deposited under  $E_V$  of 269 J/mm<sup>3</sup>.

The difference of mechanical properties is significant along directions between parallel and perpendicular to  $c$ -axis for HCP materials [10]. In this study, two typical orientations, i.e.  $\{11\bar{2}0\} \langle 1\bar{1}02 \rangle$  prismatic orientation whose  $c$ -axis aligns perpendicular to  $Z$ -axis and  $\{0002\} \langle 1\bar{1}00 \rangle$  basal orientation whose  $c$ -axis is parallel to  $Z$ -axis, are defined as orientation I (Fig. 8(a, c)) and orientation II (Fig. 8(b, d)), respectively. Accordingly, the loading force perpendicular and parallel to  $c$ -axis of orientation is also defined as stress states I (Fig. 8(a, d)) and II (Fig. 8(b, c)), respectively.

The different build orientations between the horizontally and vertically built tensile samples would result in the different loading force directions during tensile test, as shown in Fig. 2. As for vertically built tensile sample, the force direction is parallel to  $Z$ -axis, resulting in that grains with orientations I and II would be in stress states I (Fig. 8a) and II (Fig. 8b), respectively. Conversely, the force direction is perpendicular to  $Z$ -axis in the horizontally built tensile samples, thus grains with orientations I and II will be in stress states II (Fig. 8c) and I (Fig. 8d), respectively. Table 2 gives the stress states for orientations I and II in vertically and horizontally built samples.

In order to evaluate the difference of their contents, orientation ratio ( $R$ ) is defined as the ratio of the contents of the two orientations, which can be calculated by the following equation:

$$R = \frac{f_1}{f_2} \quad (1)$$

where  $f_1$  and  $f_2$  refer to the volume fractions of orientations I and II respectively. The volume fractions of orientations I and II as well as the calculated  $R$  value of the SLMed Ti-6Al-4V samples deposited under various energy densities are given in Table 3. Clearly, with increasing laser energy density, the volume fraction of orientation I ( $f_1$ ) increases from 3.73% to 13.15%, while the volume fraction of orientation II ( $f_2$ ) reduces from 1.97% to 0.79%, resulting in an increase of  $R$  value from 1.89 to 16.65.

According to the Burgers orientation relationship, there are 12 equivalent crystallographic variants within each prior  $\beta$  grain during phase transformation of  $\beta \rightarrow \alpha'$  [16]. As a result,  $\alpha'/\alpha'$  boundaries within a prior  $\beta$  grain form by the combinations of any two of these equivalent variants. Furthermore, there will be five kinds of high angle grain boundaries (HAGBs), which are  $10^\circ$ ,  $60^\circ$ ,  $61^\circ$ ,  $63^\circ$  and  $90^\circ$  with probabilities of 18.2%, 36.4%, 18.2%, 18.2% and 9.0%, respectively [25]. It is easy to find that the probability of HAGBs with  $\sim 60^\circ$  can even reach 72.8%. Fig. 9 shows the distribution of misorientation angle in the SLMed Ti-6Al-4V samples deposited under various laser energy densities. A dominated fraction of HAGBs ( $>80\%$ ) further implies the weak martensite texture, as mentioned above. Expectedly, an obvious peak located around  $60^\circ$  also demonstrates the existence of Burgers orientation relationship in the SLMed samples. However, the average fraction of HAGBs with  $\sim 60^\circ$  is 41.3%, which is much less than the theoretical value (72.8%). Therefore, it can be deduced that a part of  $\beta \rightarrow \alpha'$  phase transformation does not follow the Burgers orientation relationship due to the extremely rapid cooling rate during SLM, which is also reported by Simonelli et al. [18].

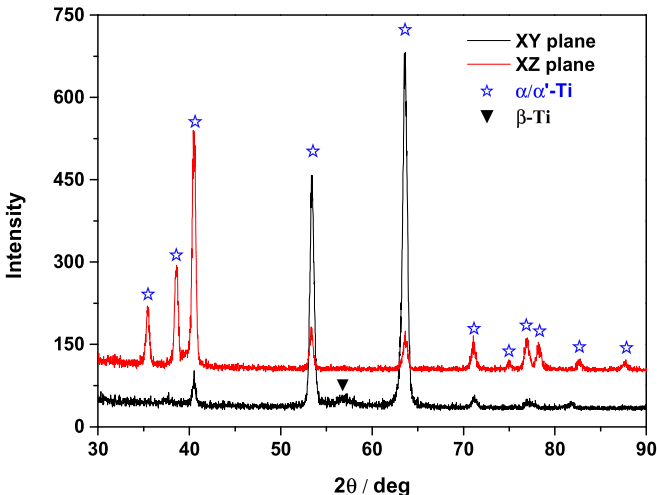
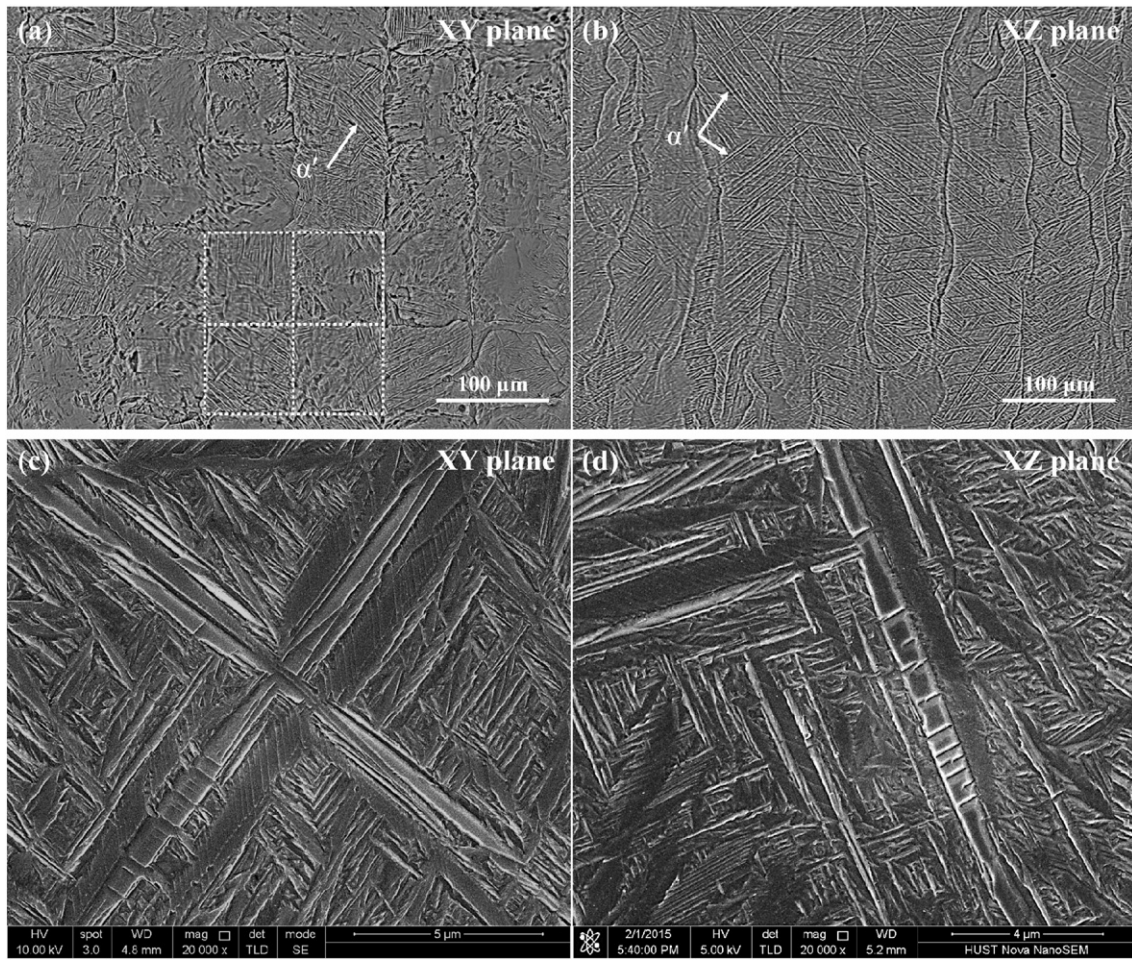
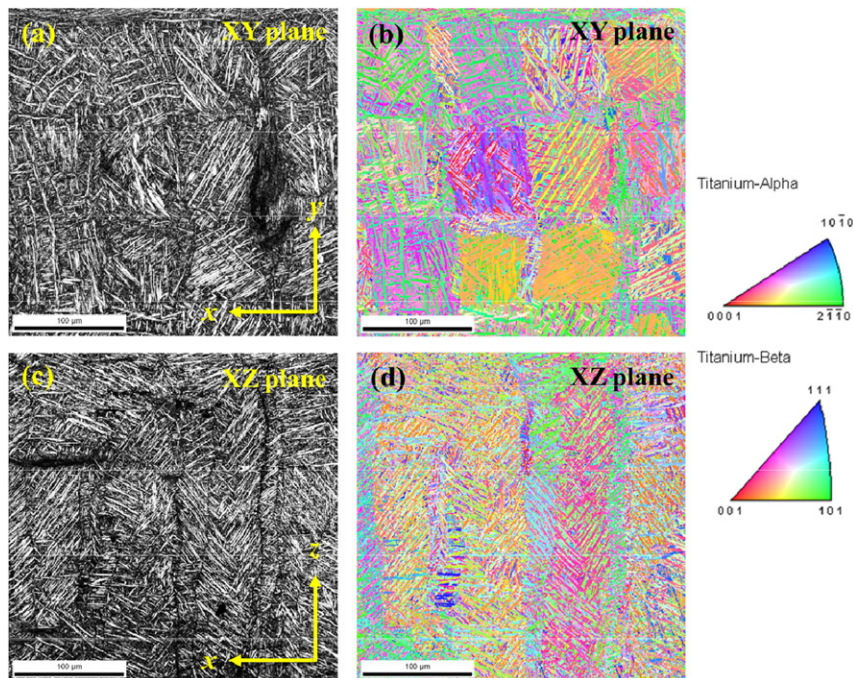


Fig. 3. X-ray diffraction patterns of the SLMed Ti-6Al-4V samples on XY and XZ planes.





**Fig. 4.** Microstructures of the SLMed Ti-6Al-4V sample under  $E_V$  of  $121 \text{ J/mm}^3$  on XY (a and c) and XZ (b and d) planes; (c) and (d) are the magnifications of the areas in (a) and (b), respectively.



**Fig. 5.** Image quality photos (a and c) and orientation images (b and d) on XY (a and b) and XZ (c and d) planes of SLMed Ti-6Al-4V sample deposited with  $121 \text{ J/mm}^3$ .



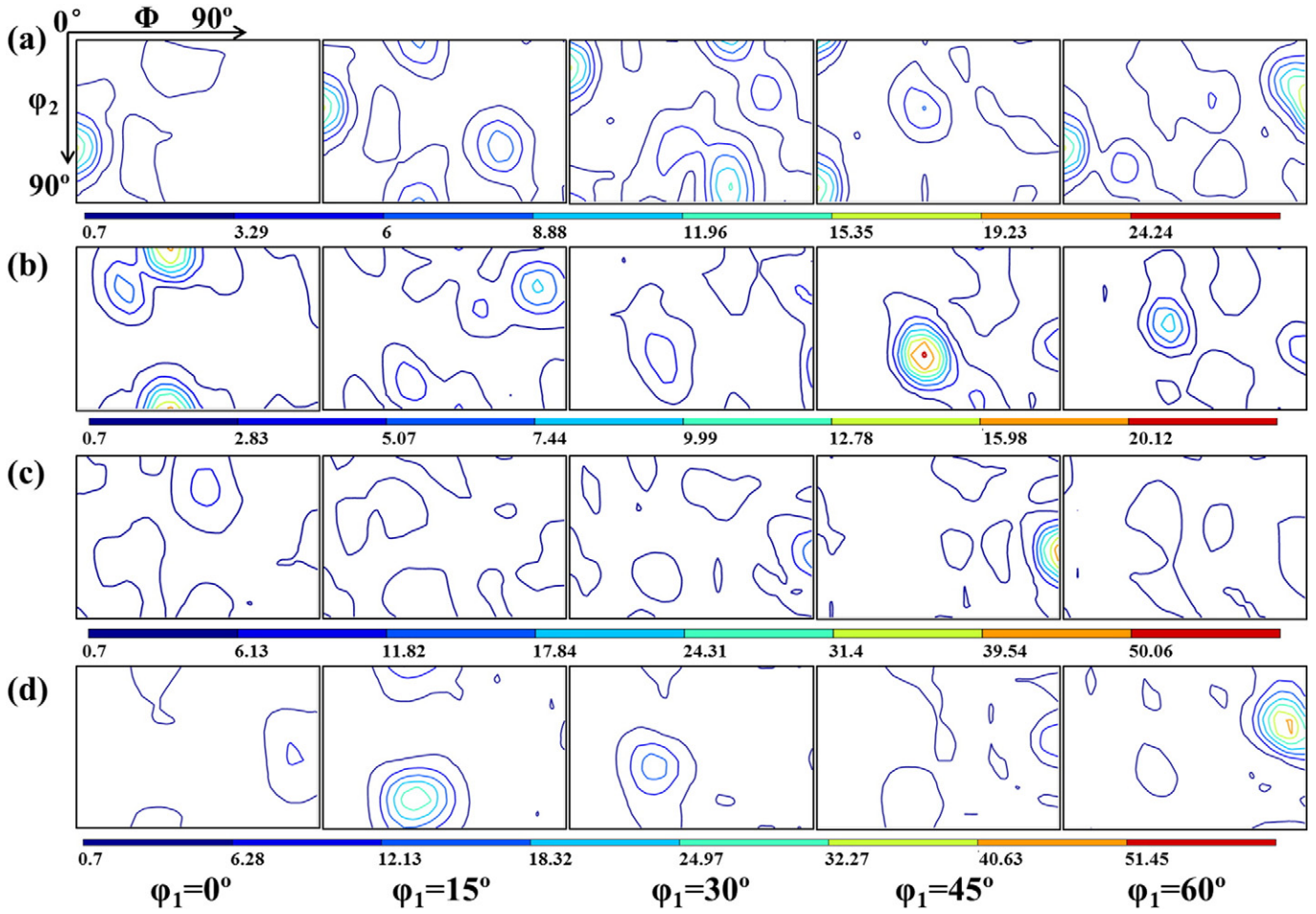


Fig. 6.  $\phi_2 = 0\text{--}60^\circ$  ODF sections of SLMed Ti-6Al-4V samples deposited under  $E_v$  of 101 J/mm<sup>3</sup> (a), 121 J/mm<sup>3</sup> (b), 230 J/mm<sup>3</sup> (c) and 269 J/mm<sup>3</sup> (d).

### 3.3. Anisotropy in tensile properties

The anisotropy in tensile properties of the SLMed Ti-6Al-4V samples is investigated in this section due to the inherently low symmetry of hexagonal close-packed (HCP) martensite  $\alpha'$  [10]. TDR (Tensile strength Difference Ratio), YDR (Yield strength Difference

Ratio) and EDR (Elongation Difference Ratio) are defined as the difference ratio between vertically and horizontally built samples to evaluate their anisotropy, which can be calculated by the following equations:

$$TDR = \frac{|TS_h - TS_v|}{TS_v} \quad (2)$$

$$YDR = \frac{|YS_h - YS_v|}{YS_v} \quad (3)$$

$$EDR = \frac{|El_h - El_v|}{El_v} \quad (4)$$

where  $TS_h$ ,  $TS_v$ ,  $YS_h$ ,  $YS_v$ ,  $El_h$  and  $El_v$  refer to the tensile and yield strengths and elongations of horizontally and vertically built Ti-6Al-4V samples, respectively.

The tensile properties of SLMed Ti-6Al-4V samples deposited along horizontal and vertical directions under various  $E_v$  are exhibited in Fig. 10. Under the laser interactions with materials [26], nearly all the tensile strengths of SLMed Ti-6Al-4V samples are superior to those of the as-cast and wrought ones [5]. However, the elongations of SLMed Ti-6Al-4V samples are fairly lower, compared with the as-cast or wrought ones. Obviously, there is an upward trend in the tensile ( $\sim 1000$  to  $\sim 1200$  MPa) and yield ( $\sim 800$  to  $\sim 1000$  MPa) strengths, and elongation ( $\sim 3.7$  to  $\sim 6.0$ ) of SLMed samples with increasing laser energy density from 101 to 230 J/mm<sup>3</sup>. The improved strengths and ductility may be ascribed to the refined microstructure of the SLMed Ti-6Al-4V samples under higher  $E_v$ . It is noteworthy that the horizontally built

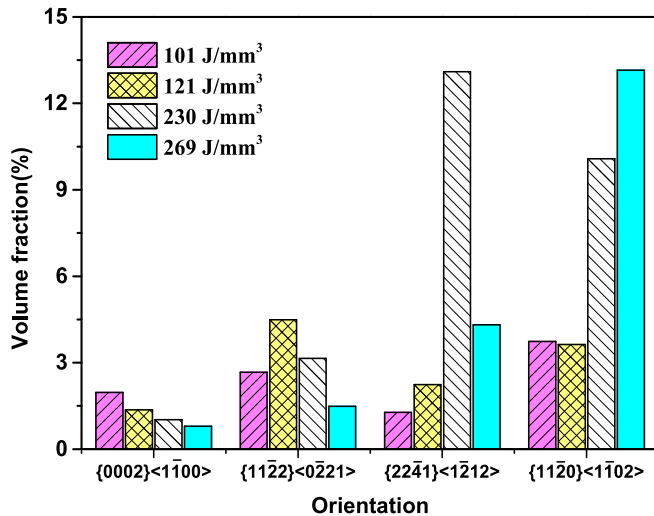
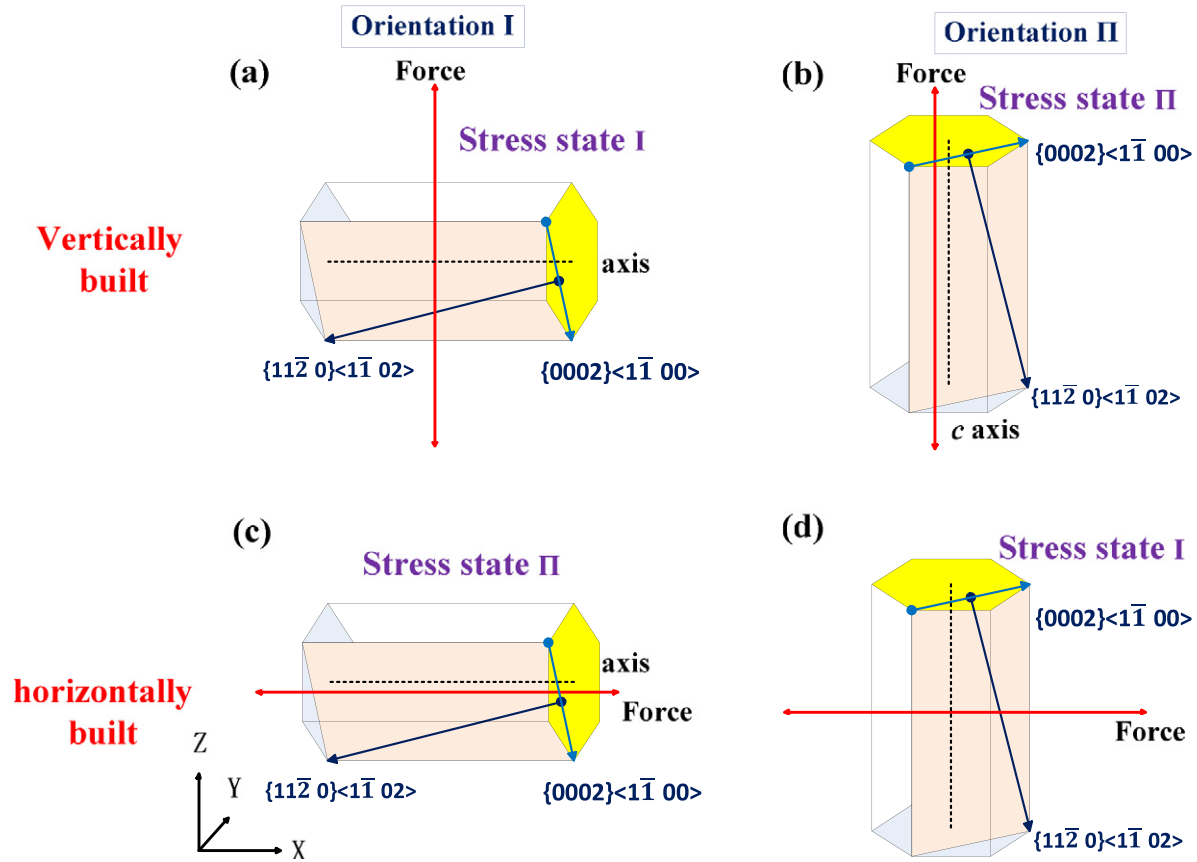


Fig. 7. The volume fractions of several types of orientations in the SLMed Ti-6Al-4V samples deposited under various  $E_v$ .



**Fig. 8.** The stress states I (a, d) and II (b, c) of grains with crystallographic orientations I (a, c) and II (b, d) in the vertically (a, b) and horizontally (c, d) built tensile samples.

Ti-6Al-4V samples produce higher strengths while the vertically built ones acquire superior elongations, suggesting that the mechanical anisotropy exists in the SLMed Ti-6Al-4V samples. Moreover, TDR, YDR and EDR also increase from 6.7% to 8.8%, 7.4% to 12.2% and 2.6% to 18.6% respectively with increasing laser energy density.

The different force loading directions may induce different stress states between the horizontally and vertically built samples. Upon load, grains with orientations I and II will be in stress states I and II for vertically-built tensile samples, respectively. Conversely, the loading force causes the grains with orientations I and II to be in stress states II and I for horizontally-built tensile samples, respectively. According to Table 3, the volume fraction of orientation I is always larger than that of orientation II in SLMed Ti-6Al-4V samples, implying that orientation I plays a more dominant role in determining the crystallographic orientation, compared with orientation II. Therefore, more grains are in stress state I in vertically-built tensile samples, compared with horizontally-built tensile samples.

The plastic deformation ability of material can be characterized by Schmid factor, which is the ratio of the maximum resolved shear stress to the maximum contact pressure. The larger the Schmid factor, the larger the resolved shear stresses will be and the more easily the slip

system can be activated [27]. Thus, Schmid factor ( $m$ ) can be determined by stress state according to the following equation [28]:

$$m = \cos\theta \cdot \cos\lambda \quad (5)$$

where  $\theta$  is the angle between the force loading direction and the normal of slip plane, and  $\lambda$  refers to the angle between the force loading direction and the slip direction.

The Schmid factor values for several typical slip systems under different stress states are different, as shown in Table 4. The Schmid factor is calculated to be zero for basal slip system. As the grains are in stress state I, Schmid factor can achieve the maximum value of 0.433 for  $\{10\bar{1}0\} \langle 11\bar{2}0 \rangle$  prismatic slip system. When the grains are in state II,  $\{10\bar{1}1\} \langle 11\bar{2}3 \rangle$  pyramidal slip system has the highest Schmid factor of 0.327. Clearly, the Schmid factor values are larger in stress state I than those in state II for  $\{10\bar{1}0\} \langle 11\bar{2}0 \rangle$  prismatic slip system and  $\{10\bar{1}1\} \langle 11\bar{2}0 \rangle$  pyramidal slip system. Also, Schmid factor values are similar in stress states of I and II for  $\{10\bar{1}1\} \langle 11\bar{2}3 \rangle$  pyramidal slip system. Thus, the grains in stress state I are easier to slip than those in state II. As mentioned above, more grains are in state I upon loading for the vertically-built tensile samples than the horizontally-built ones.

**Table 2**  
Stress state for orientations I and II in vertically and horizontally built samples.

| Orientation type | Stress state             |                            |
|------------------|--------------------------|----------------------------|
|                  | Vertically-built samples | Horizontally-built samples |
| Orientation I    | Stress state I           | Stress state II            |
| Orientation II   | Stress state II          | Stress state I             |

**Table 3**  
The volume fractions of orientations I and II in SLMed Ti-6Al-4V samples under various energy laser densities.

| Type                  | $E_V = 101$<br>J/mm <sup>3</sup> | $E_V = 121$<br>J/mm <sup>3</sup> | $E_V = 230$<br>J/mm <sup>3</sup> | $E_V = 269$<br>J/mm <sup>3</sup> |
|-----------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Orientation I, $f_1$  | 3.73%                            | 3.63%                            | 10.07%                           | 13.15%                           |
| Orientation II, $f_2$ | 1.97%                            | 1.36%                            | 1.02%                            | 0.79%                            |
| R                     | 1.89                             | 2.67                             | 9.87                             | 16.65                            |

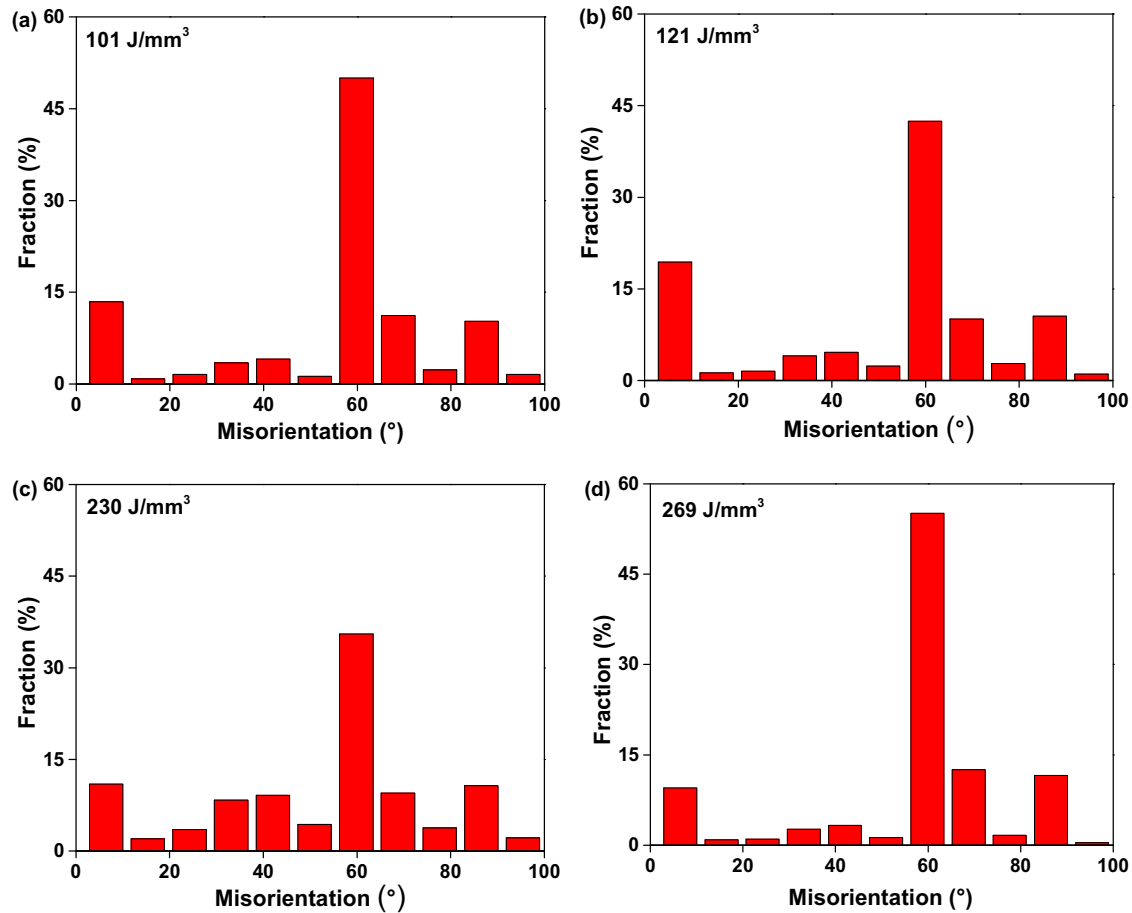


Fig. 9. The misorientation distribution of misorientation angle in SLMed Ti-6Al-4V deposited under various  $E_V$ .

Consequently, the high elongation in the vertically-built tensile samples may be attributed to their higher Schmid factor values. Besides, the growing  $R$  value (from 0.89 to 15.64) with laser energy density results from the fact that the volume fraction difference between grains in state I and state II becomes larger. Therefore, the anisotropy in tensile properties of the SLMed Ti-6Al-4V samples becomes stronger with laser energy density. Thus, adjusting processing parameters to reduce laser energy density can weaken the anisotropy of the SLMed Ti-6Al-4V samples.

The fracture surface morphologies of the SLMed Ti-6Al-4V tensile samples deposited vertically and horizontally under various  $E_V$  are exhibited in Fig. 11. The existence of a large number of dimples indicates that the fracture mode of the SLMed Ti-6Al-4V samples is dominated by ductile fracture. However, most of the dimples with diameters of about 1–5  $\mu\text{m}$  are small and shallow, implying the relatively low ductility of the SLMed Ti-6Al-4V samples. With increasing  $E_V$  from 101 to 230 J/mm³, average size of dimples in horizontally and vertically built samples increases from ~2 to 8  $\mu\text{m}$  and ~7 to 20  $\mu\text{m}$ , respectively.

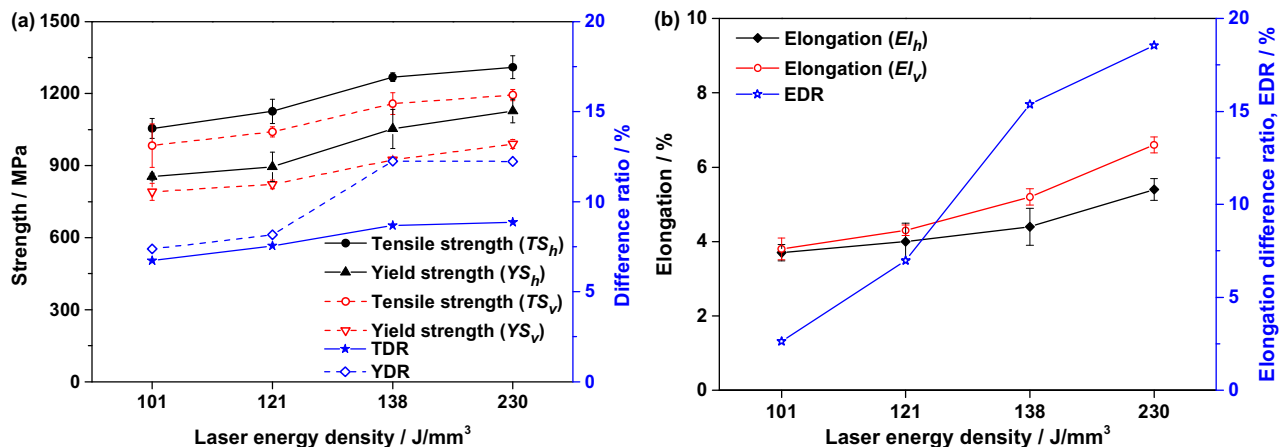


Fig. 10. The tensile and yield strengths (a) and elongation (b) of the SLMed Ti-6Al-4V samples under various laser energy densities.

**Table 4**  
Schmid factor for several typical slip systems under different stress states.

| Slip system                                                                        | Stress state I | Stress state II |
|------------------------------------------------------------------------------------|----------------|-----------------|
| Basal $\langle a \rangle$ slip $\{0002\} \langle 11\bar{2}0 \rangle$               | 0              | 0               |
| Prismatic $\langle a \rangle$ slip $\{10\bar{1}0\} \langle 11\bar{2}0 \rangle$     | 0.433          | 0               |
| Pyramidal $\langle a \rangle$ slip $\{10\bar{1}1\} \langle 11\bar{2}0 \rangle$     | 0.224          | 0               |
| Pyramidal $\langle a + c \rangle$ slip $\{11\bar{2}2\} \langle 11\bar{2}3 \rangle$ | 0.286          | 0.327           |

Apparently, the size of dimples in horizontally-built ones (Fig. 11(a–c)) is smaller (Fig. 11(d–f)) than that in vertically-built samples, resulting in that the vertically-built samples have a better ductility.

#### 4. Conclusion

The crystallographic texture and mechanical anisotropy of the SLMed Ti-6Al-4V samples under various laser energy densities were investigated in the present work. The major conclusions are listed as follows:

- (1) The microstructure of SLMed Ti-6Al-4V samples is composed of martensites within columnar prior  $\beta$  grains, showing a weak texture of  $\alpha'$ .
- (2) As laser energy density increases from 101 to 269 J/mm<sup>3</sup>, the dominated crystallographic orientation of martensites transforms in the order:  $(12\bar{3}0)[2\bar{1}13] \rightarrow (11\bar{2}4)[\bar{1}341] \rightarrow (11\bar{2}0)[1\bar{1}01] \rightarrow (11\bar{2}0)[2\bar{2}03]$ . Also, the contents of  $\{11\bar{2}0\} \langle 1\bar{1}02 \rangle$  prismatic orientation are always larger than those of  $\{0002\} \langle 1\bar{1}00 \rangle$  basal orientation for martensites in the SLMed Ti-6Al-4V samples. The fraction of  $\{0002\} \langle 1\bar{1}00 \rangle$  basal orientation

decreases while the fraction of  $\{11\bar{2}0\} \langle 1\bar{1}02 \rangle$  prismatic orientation increases with laser energy density. As a result, the difference ratio between them increases from 0.89 to 15.64 as the applied  $E_v$  increases from 101 to 269 J/mm<sup>3</sup>.

- (3) The horizontally built Ti-6Al-4V samples possess higher strengths, but the vertically built ones get better elongations, verifying the existence of mechanical anisotropy in SLMed Ti-6Al-4V samples. Moreover, the anisotropy in tensile properties of the SLMed Ti-6Al-4V samples can be weakened by reducing laser energy density.
- (4) The Schmid factor values of the grains in the vertically-built tensile samples are generally higher than those in the horizontally-built ones, which is closely associated with the formation of anisotropy in SLMed Ti-6Al-4V samples.

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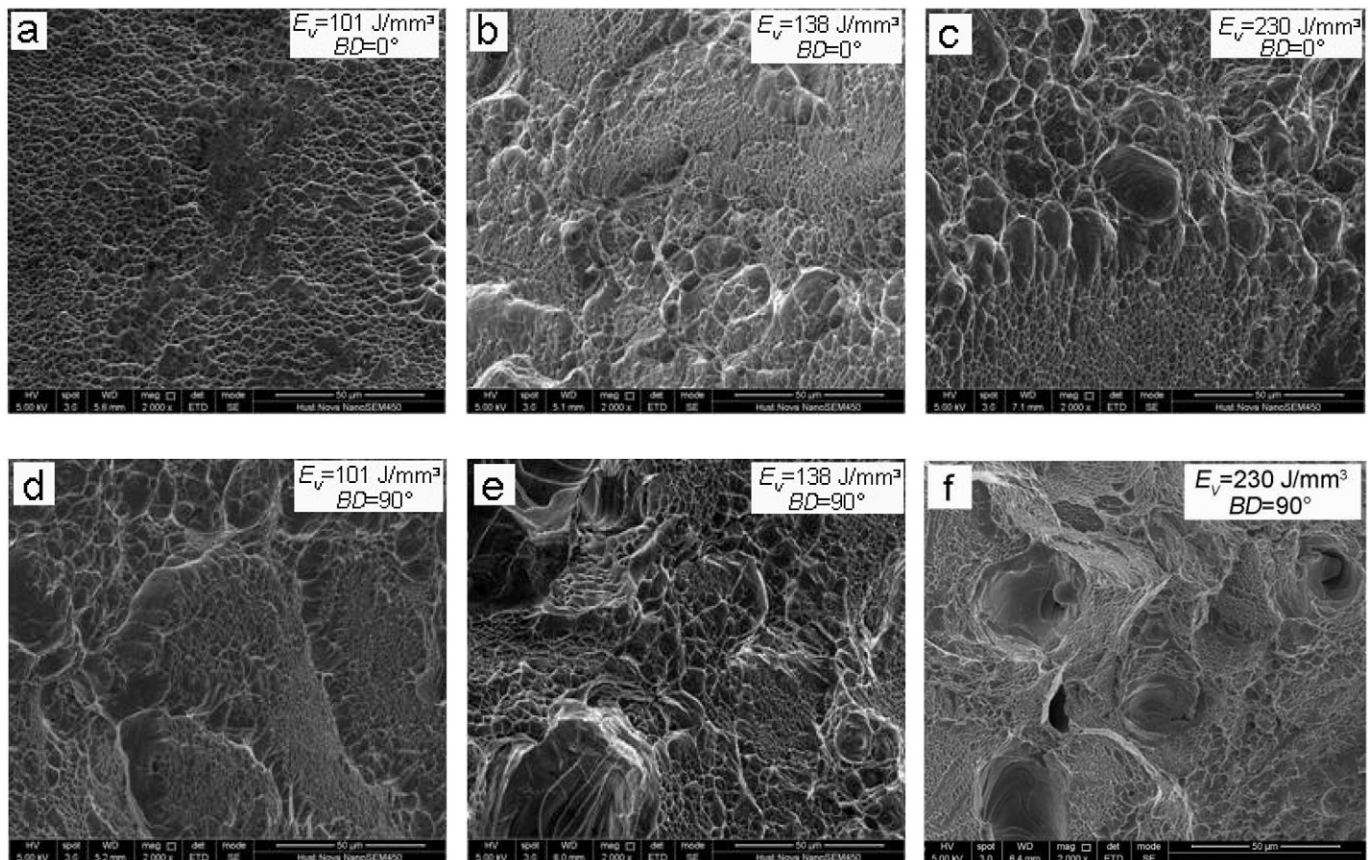


Fig. 11. Fracture morphologies of the SLMed Ti-6Al-4V tensile samples.



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